On Miller's planet (Chapter 17) the tidal forces are enormously larger and are key to the huge waves that Cooper and his crew encounter.

We now have three points of view on tidal forces:

- *Newton's viewpoint* (Figure 4.8): The Earth does not feel the Moon's full gravitational pull, but rather the full pull (which varies over the Earth) minus the average pull.
- *The tendex viewpoint* (Figure 4.9): The Moon's tendex lines stretch and squeeze the Earth's oceans; also (Figure 4.7) a black hole's tendex lines stretch and squeeze the paths of planets and stars around the black hole.
- *The straightest-route viewpoint* (Figure 4.6): The paths of stars and planets around a black hole are the straightest routes possible in the hole's warped space and time.

Having three different viewpoints on the same phenomenon can be extremely valuable. Scientists and engineers spend most of their lives trying to solve puzzles. The puzzle may be how to design a spacecraft. Or it may be figuring out how black holes behave. Whatever the puzzle may be, if one viewpoint doesn't yield progress, another viewpoint may. Peering at the puzzle first from one viewpoint and then from another can often trigger new ideas. This is what Professor Brand does, in *Interstellar*, when trying to understand and harness gravitational anomalies (Chapters 24 and 25). This is what I've spent most of my adult life doing.